https://github.com/Harsh-IIT-Ropar/insurance-cost-prediction

Prediction of Annual Medical Insurance Cost Using Linear Regression

HARSH RAJ - Documentation Lead
PREM KIRAN GEDALA - PPT Lead
RAKSHITH A Y - Video Lead
NIKHIL MEENA - Execution Lead

Problem and data

This dataset is a realistic synthetic healthcare insurance dataset created for machine learning, data science, healthcare analytics, and AI-based prediction projects.
It contains demographic, lifestyle, medical history, chronic disease, hospitalization, and insurance-related information of individuals along with their annual medical costs.

The dataset is specially designed for:

- *Medical Insurance Cost Prediction*
- *Healthcare Analytics*
- *Risk Assessment Systems*
- *Regression Machine Learning Projects*
- *Predictive Modeling*
- *AI Healthcare Dashboards*
- *Data Visualization & EDA*
- *Feature Engineering Practice*

📊 *Dataset Overview*

- *Rows: 7,500*
- *Columns: 22*
- *Target Variable: annual_medical_cost_usd*
- *Dataset Type: Structured Tabular Data*
- *ML Problem Type: Regression*

Project Pipeline Diagram

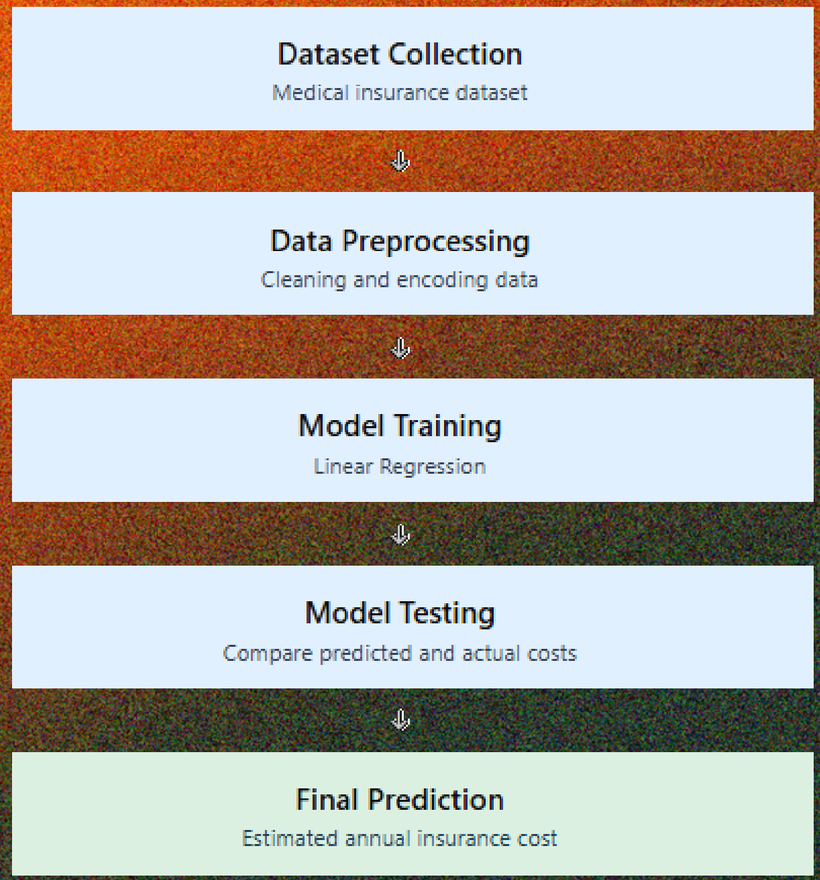
* *Approach Used – Linear Regression*

Linear Regression is a supervised Machine Learning algorithm used to predict continuous numerical values.
In our project, it predicts annual medical insurance charges based on different input factors such as age, BMI, smoking habits and other health-related details.

* *Linear Regression Formula:*
$$Y=b_0+b_1X_1+b_2X_2+\cdots+b_nX_n$$

Where:

- Y = *Predicted insurance cost.*
- b_0 = *Intercept.*
- $b_1,b_2,\ldots,b_n$ = *Model coefficients.*
- $X_1,X_2,\ldots,X_n$ = *Input features.*

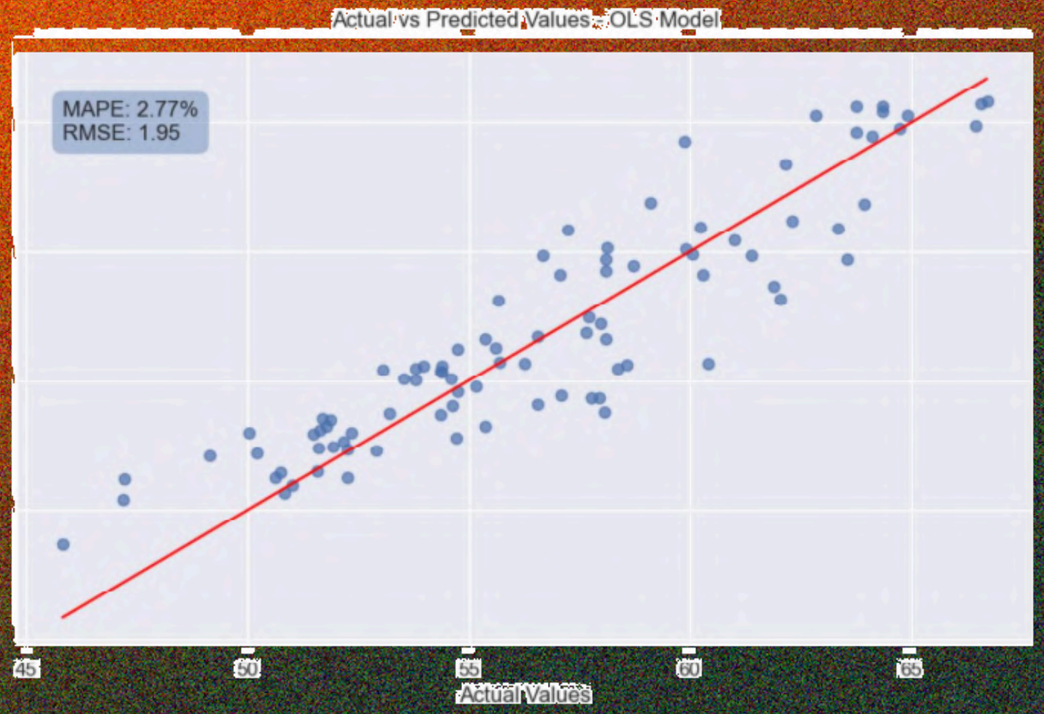


Result of the Project

* After training and testing the Linear Regression model, we obtain predicted medical insurance costs based on the given input information.

Important Performance Metrics
We evaluate the model using the following metrics:
R^2 Score: Measures how well the model explains variations in insurance costs. A value closer to 1 generally indicates a better fit.

Mean Absolute Error (MAE): Measures the average absolute difference between actual and predicted insurance costs.
Mean Squared Error (MSE): Measures the average squared prediction error.



* *Supports Public Hygiene*

Proper sanitation facilities are essential for community health and must be integrated into health care systems.

Learnings

1. What Worked?

- Linear Regression successfully predicts numerical insurance costs.
- Personal and lifestyle information helps estimate insurance expenses.
- Pandas made data processing easier.
- Matplotlib helped visualize the results.

2. What Didn't Work?

- Prediction errors may occur because of differences between actual and predicted values.
- Linear Regression may not capture all complex relationships in medical insurance costs.
- Missing or incorrect data can affect model performance.

3. AI- Use Disclosure

AI tools were used to support project understanding, code assistance and presentation preparation. The dataset processing, model implementation and result analysis were performed as part of our project work.

thank you

Building a Healthier Tomorrow